

Instructions

- Answer all eight questions. Show sufficient working for method marks.
- A calculator may be used unless a question states otherwise.
- Write clearly in the spaces provided. Additional pages may be attached.

SCORE SAVED TO COURSE PROGRESS

36 / 40

1

Convert the binary number to denary and explain the place-value method used.

[4 marks]

 $1011\ 0110_2$ *Method sound; final simplification omitted.*

4/4

2

Describe how a hash table resolves collisions, naming one strategy and its drawback.

[6 marks]

Check the condition before applying the rule.

6/6

AGENT ANNOTATED

3

Write a function that returns the maximum value in a list without using a built-in max.

[5 marks]

Correct and clearly justified.

4/5

4

Explain why binary search requires a sorted list, and state its complexity.

[5 marks]

 $O(\log n)$

Method sound; final simplification omitted.

4/5

5

Compare compiled and interpreted execution, giving one advantage of each.

[5 marks]

Check the condition before applying the rule.

5/5

AGENT ANNOTATED

6

Describe the purpose of a primary key in a relational database and why duplicates break it.

[4 marks]

Correct and clearly justified.

3/4

7

Explain recursion using a worked example, and state the role of the base case.

[6 marks]

Method sound; final simplification omitted.

6/6

8

Identify the logic error in the described loop and explain the correction.

[5 marks]

Check the condition before applying the rule.

4/5